

EC824B

Ridge regression and LASSO

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Ridge regression and LASSO

- Two popular methods for supervised dimension reduction

$$Y = X'\beta + \varepsilon,$$

where $X \in \mathbb{R}^{N \times K}$ where $K \gg N$

- LASSO: ℓ_1 -norm penalised regression, for appropriate λ_1

$$\left\{ \operatorname{argmin}_{\beta \in \mathbb{R}^K} \frac{1}{2N} \|Y - X\beta\|^2 \text{ s.t. } \|\beta\|_1 \leq R_1 \right\} \equiv \left\{ \operatorname{argmin}_{\beta \in \mathbb{R}^K} \frac{1}{2N} \|Y - X\beta\|^2 + \lambda_1 \|\beta\|_1 \right\}$$

- Ridge regression: ℓ_2 -norm penalised regression, for appropriate λ_2

$$\left\{ \operatorname{argmin}_{\beta \in \mathbb{R}^K} \frac{1}{N} \|Y - X\beta\|^2 \text{ s.t. } \|\beta\|_2^2 \leq R_2 \right\} \equiv \left\{ \operatorname{argmin}_{\beta \in \mathbb{R}^K} \frac{1}{N} \|Y - X\beta\|^2 + \lambda_2 \|\beta\|_2^2 \right\}$$

Shrinkage

The LASSO and Ridge regression solutions imply “shrinking” the coefficients

Why might we want to use shrinkage in a linear model?

- Gauss-Markov tells us OLS is the lowest variance estimator in the class of linear *unbiased* estimators
- What if we are willing to give up some bias to decrease variance?
- Remember $MSE(\hat{\beta}) = \mathbb{E}[(\hat{\beta} - \beta)^2] = Bias(\hat{\beta})^2 + Var(\hat{\beta})$
- If we want to minimise MSE , is it optimal to give up some $Bias$ to reduce variance?
- This is known as the bias variance trade-off, and becomes more important when the number of parameters K is the same or larger order than the sample size N
- In fact, for $K > N$, dimension reduction is required for identification

1 Best subset (ℓ_0 penalty)

2 LASSO (ℓ_1 penalty)

3 Ridge (ℓ_2 penalty)

Best subset (ℓ_0 penalty)

Let's start with an infeasible problem statement. We wish to find the best subset of regressors from a large set of covariates:

$$\min_{\beta \in \mathbb{R}^K} \frac{1}{2} \|Y - X\beta\|^2 \quad \text{s.t.} \quad \sum_{k=1}^K \mathbb{1}\{\beta_k \neq 0\} \leq K_0$$

This is equivalent to restricting the 0 -“norm” of the set of coefficients,

$$\|\beta\|_0 := \sum_k \beta^0 := \sum_k \mathbb{1}\{\beta_k \neq 0\}^1$$

- Infeasible: this problem is non-convex and combinatorial(ly hard)
- 0 -“norm” is non-convex, no dual Lagrangian: the problem cannot be simplified!

¹For this “norm” use as definition that $0^0 = 0$

Best subset (ℓ_0 penalty) - non-convex

Remember from PCA slides what it means to be non-convex for sets:

$$\begin{aligned} \mathcal{S} &= \{S : \text{condition on } S \text{ satisfied}\} \\ \text{if } S_1, S_2 &\in \mathcal{S} \quad \text{and} \quad \exists \alpha \in [0, 1] \\ \text{s.t. } \alpha S_1 + (1 - \alpha) S_2 &\notin \mathcal{S} \quad \Rightarrow \quad \mathcal{S} \text{ non-convex set.} \end{aligned}$$

Then take the set of β after applying our $\|\beta\|_0$ penalty, e.g.,

$$\begin{aligned} \mathcal{S} &= \{S : \|S\|_0 \leq 2\} \\ \Rightarrow \mathcal{A} &= \{\beta_1, 0, \beta_3\} \in \mathcal{S}, \quad \mathcal{B} = \{0, \beta_2, \beta_3\} \in \mathcal{S} \\ \text{but } \alpha \mathcal{A} + (1 - \alpha) \mathcal{B} &= \{\alpha \beta_1, (1 - \alpha) \beta_2, \beta_3\} \notin \mathcal{S} \quad \forall \alpha \in (0, 1) \end{aligned}$$

In general, the 0-norm penalty is non-convex, which makes computation difficult

Best subset (ℓ_0 penalty) - combinatorial(ly hard)

We must “manually” try every combination of subsets:

$$\text{For } \|\beta\|_0 \leq K_0 \quad \text{check} \quad \sum_{j=1}^{K_0-1} \binom{K}{j}$$

which grows exponentially in K_0

- Hence, need to consider convex relaxations of this problem

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LASSO (ℓ_1 penalty)

Consider instead of the ℓ_0 -“norm”, the ℓ_1 -norm, that is actually a norm,

$$\|\beta\|_1 = |\beta_1| + |\beta_2| + \cdots + |\beta_K|$$

and compare the constraints,

$$\|\beta\|_0 \leq K_0 \quad \text{vs.} \quad \|\beta\|_1 \leq R_1$$

- Right side is now convex!
- But does solving with that constraint always satisfy the left hand side? I.e. will the problems be dual (equivalent)?
- Unfortunately not (failure of Lagrangian duality for non-convex constraint), but there are some instances where they can have close solutions
- Least Absolute Shrinkage and Selection Operator

LASSO (ℓ_1 penalty)

Convex:

$$\begin{aligned} \|A\|_1 \leq R_1 \quad \& \quad \|B\|_1 \leq R_1 \\ \Rightarrow \|\alpha A + (1 - \alpha)B\|_1 &\leq \|\alpha A\|_1 + \|(1 - \alpha)B\|_1 \\ &= \alpha\|A\|_1 + (1 - \alpha)\|B\|_1 \leq R_1 \end{aligned}$$

The inequality on the second line is tight when A and B are orthogonal, so when we take combinations of objects with linearly related components, the penalty can be slack:

$$\begin{aligned} A &= \{\beta_1, 0\}, \quad B = -\{b\beta_1, (1 - b)\beta_2\}, \quad \text{for } b \in [0, 1] \\ \Rightarrow \|A\|_1 &= |\beta_1|, \quad \|B\|_1 = b|\beta_1| + (1 - b)|\beta_2| \\ \text{and } \|\alpha A + (1 - \alpha)B\|_1 &= |\alpha - (1 - \alpha)b||\beta_1| + (1 - \alpha)(1 - b)|\beta_2| \\ &\xrightarrow{b \rightarrow 1} |2\alpha - 1||\beta_1| \quad \text{and} \quad \xrightarrow{b \rightarrow 0} \alpha\|A\|_1 + (1 - \alpha)\|B\|_1 \end{aligned}$$

Hence, penalty can be slackened for sets with linear dependence

LASSO (ℓ_1 penalty)

Why is this useful to know?

- Usually the unrestricted optimum is outside the feasible set $\|\beta\|_1 \leq R_1$
- Hence, the restricted solution is usually on the boundary $\|\beta\|_1 = R_1$

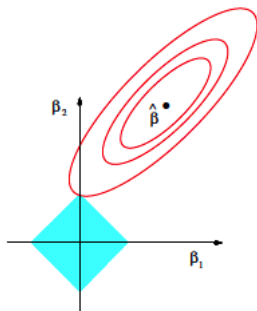


Figure: Blue area is feasible set, $\|\beta\|_1 \leq R_1$. Red lines are “level sets” of $\|Y - X\beta\|^2$ around the optimal unrestricted least squared (OLS) value, labelled $\hat{\beta}$.

LASSO (ℓ_1 penalty) - Solution

Solution (normalise X s.t. $\frac{1}{n}X'X = \mathbb{I}_K$):

$$\hat{\beta}_k = \begin{cases} \hat{\beta}_{OLS,k} - \lambda_1 \text{sign}(\hat{\beta}_{OLS,k}) & \text{if } |\hat{\beta}_{OLS,k}| \geq \lambda_1 \\ 0 & \text{otherwise} \end{cases}$$

or in shorthand, $\hat{\beta}_k = \text{sign}(\hat{\beta}_{OLS,k})(|\hat{\beta}_{OLS,k}| - \lambda_1)_+$. To see this, note the FOC when $\beta_k \neq 0$

$$\text{FOC} = \begin{cases} -\hat{\beta}_{OLS,k} + \beta_k + \lambda_1 & \text{if } \beta_k > 0 \\ -\hat{\beta}_{OLS,k} + \beta_k - \lambda_1 & \text{if } \beta_k < 0 \end{cases}$$

In the problem set you will show why $\hat{\beta}_k = 0$ when $|\hat{\beta}_{OLS,k}| < \lambda_1$, and prove the result more generally.

LASSO (ℓ_1 penalty) - Hyperparameter optimisation

The hyperparameter λ_1 cannot be explicitly optimised in the solution concept. Optimal λ_1 depends on objective. For a prediction problem, we want to find λ_1 such that given a new observation X_{N+1} , want smallest MSE for $\hat{Y}_{N+1} = X'_{N+1}\hat{\beta}$.

$$\min_{\lambda > 0} \mathbb{E} \left(Y_{N+1} - X'_{N+1} \hat{\beta}(\lambda) \right)^2$$

where $\hat{\beta}(\lambda)$ is the LASSO optimal for $\lambda_1 = \lambda$

- Use *cross-validation* to numerically solve this, making use of *iid* sampling assumption:

$$\lambda_1^* = \operatorname{argmin}_{\lambda > 0} \sum_i \left(Y_i - X'_i \hat{\beta}^{(-i)}(\lambda) \right)^2$$

where $\hat{\beta}^{(-i)}(\lambda)$ is the estimate when dropping i when estimating $\hat{\beta}(\lambda)$

LASSO (ℓ_1 penalty) - Forecasting application

Say we are interested solely in forecasting Y with a large set of $X_i \in \mathbb{R}^K$:

$$\frac{1}{N} \|X\beta - X\hat{\beta}(\lambda_1^*)\|^2 \leq C\|\beta\|_1 \sqrt{\frac{\log(K)}{N}}$$

see [Hastie, Tibshirani and Wainwright (2015); Wainwright (2019)]

- For this to be small, $\|\beta\|_1$ should be small
- This is a general result that allows for $K \gg N$
- If willing to impose sparsity, say $\|\beta\|_0 \leq s < K$,

$$\frac{1}{N} \|X\beta - X\hat{\beta}(\lambda_1^*)\|^2 \leq C \frac{s \log(K)}{N}$$

which converges faster

LASSO (ℓ_1 penalty) - Model selection

What if we actually want to know β ? Then we are interested in model selection:

$$\mathcal{S} := \{j \in \{1, \dots, K\} : \beta_j \neq 0\}$$

It turns out that LASSO can return the correct support:

$$\hat{\mathcal{S}} \xrightarrow{P} \mathcal{S}$$

Then, for consistency of the β estimates themselves, we can simply perform post-LASSO OLS estimation. That is, once LASSO selects $\hat{\mathcal{S}}$, we perform OLS only on the variables in $\hat{\mathcal{S}}$. The post-LASSO estimator then avoids the shrinkage bias from the λ_1 term in $\hat{\beta}_k = \text{sign}(\beta_{OLS,k})(|\beta_{OLS,k}| - \lambda_1)_+$ for the orthonormal X case.

- Look at post LASSO inference next lecture
- Inference is not as straightforward as standard OLS on the post selection covariates

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Ridge (ℓ_2 penalty)

Ridge regression: ℓ_2 -norm penalised regression

$$\min_{\beta \in \mathbb{R}^K} \frac{1}{N} \|Y - X\beta\|^2 + \lambda_2 \|\beta\|_2^2$$

- This constraint is also convex (check for yourself!)
- FOCs give the neat solution:

$$\hat{\beta}_{ridge} = (X'X + N\lambda_2 \mathbb{I}_K)^{-1} X'Y$$

- Solution does not force any coefficients to be exactly zero, which make post Ridge regression infeasible

Ridge (ℓ_2 penalty)

$$\hat{\beta}_{ridge} = (X'X + N\lambda_2\mathbb{I}_K)^{-1} X'Y$$

- $N\lambda_2\mathbb{I}_K$ regularises $X'X$, which is important when $X'X$ is close to singular (this regularisation makes the term more stable, with smaller variance)
- Again, bias added since $\mathbb{E}\hat{\beta}_{ridge} \neq \beta$, but regularisation always weakly reduces variance
- There is always a choice for λ_2 s.t. $\sum_{k=1}^K MSE(\hat{\beta}_{ridge,k}) \leq \sum_{k=1}^K MSE(\hat{\beta}_{OLS,k})$

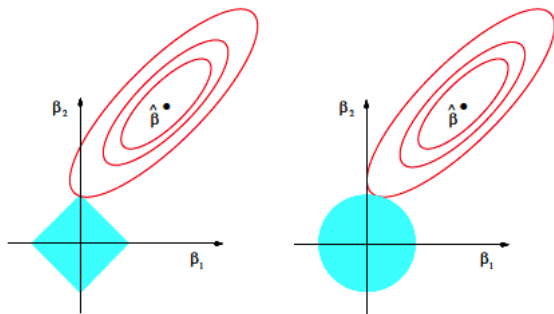
Ridge (ℓ_2 penalty)

Figure: Left is LASSO, right is Ridge. Blue area is feasible set, red lines are level sets of $\|Y - X\beta\|^2$

- LASSO is more likely to “zero” out variables, i.e. do variable selection
- Ridge only shrinks, and rarely “zeroes” out

This lecture

- Best subset is not computationally feasible: use convex relaxations
- LASSO useful for subset selection, and post-LASSO estimation can remove shrinkage bias
- Ridge also reduces MSE, but post-Ridge not feasible (aim is to regularise across all X)

Caveat: neither LASSO nor Ridge are scale-invariant

- If we increase the scale of a variable, and thus reduce the scale of its coefficient, the shrinkage is relatively higher w.r.t. λ_1 or λ_2
- For LASSO, the variable is zeroed out with higher probability
- In practice, usually standardise variables to unit variance then conduct analysis

Aside: James-Stein

The James-Stein approach first saw the opportunity to reduce MSE by giving up bias.

- If $Y \in \mathbb{R}^M \sim N(\theta, \sigma^2 \mathbb{I}_M)$ with known σ and unknown $\theta \in \mathbb{R}^M$
- OLS/MLE estimate $\hat{\theta} = Y$
- James-Stein estimate:

$$\hat{\theta}_{JS} = \left(1 - \frac{(M-2)\sigma^2}{\|Y\|^2} \right) Y$$

- This is the beginning of the use of “shrinkage”
- For $M > 2$ it can be shown that $MSE(\hat{\theta}_{JS}) < MSE(\hat{\theta})$, so $\hat{\theta}$ is *inadmissible*

References

- Hastie, T., R. Tibshirani, and M. Wainwright (2015). *Statistical learning with sparsity: the lasso and generalizations*. CRC press.
- Wainwright, M. J. (2019). *High-dimensional statistics: A non-asymptotic viewpoint*, Volume 48. Cambridge university press.